ANKIT KHUNT

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Summary

Theoretical and computational astrophysicist specializing in relativistic astrophysics, gravity, and compact objects. My research interests include neutron stars, pulsars, relativistic stellar structure, strong-field gravity, dense matter, anisotropic compact stars, dark matter in compact objects, accretion phenomena, quasi-periodic oscillations (QPOs), and modified theories of gravity. I investigate the structure, stability, oscillation properties, tidal deformability, and gravitational-wave signatures of compact objects using analytical and numerical approaches. I am particularly interested in connecting theoretical models with electromagnetic and multimessenger observations to probe the behavior of matter and spacetime under extreme conditions. My broader goal is to understand gravity, dense matter, and fundamental physics in the strong-field regime across astrophysical and cosmological scales.

Research Interests

- **Tests of Gravity with Astrophysical Compact Objects:** Neutron stars, black holes, strong-field gravity, General Relativity, and modified theories of gravity.
- **Physics of Compact Objects:** Relativistic stellar structure, anisotropic compact stars, dark matter-admixed neutron stars, dark matter capture and collapse, and black hole formation.
- **Gravitational Waves and Compact Binary Coalescence:** Compact binary mergers, quasi-normal modes, gravitational-wave echoes, and the physics and gravity of post-merger gravitational-wave signals.
- **Relativistic Asteroseismology:** Radial and non-radial oscillations, quasi-periodic oscillations (QPOs), and oscillation modes of compact stars.
- **Multi-Messenger Astrophysics:** Joint gravitational-wave, electromagnetic, and pulsar observations to constrain dense matter and gravity theories.

Education

2018–2024 **Ph.D. in Relativistic Astrophysics**
Hadron Physics Theory Lab, Department of Physics, Sardar Patel University (SPU), India
Thesis: *Study of compact stars and their properties based on general relativistic core-envelope models* (arXiv:2410.09783)
Supervisor: Prof. P. C. Vinodkumar

2016–2017 **M.Phil. in Physics**
Department of Physics, Sardar Patel University, India
Dissertation title: *Elastic properties of* $Mg_{65}Cu_{25}Tb_{10}$ *and* $Tm_{55}Al_{25}Co_{20}$ *bulk metallic glasses using pseudopotential theory*

2014–2016 **M.Sc. in Physics**
Department of Physics, Sardar Patel University, India

2011–2014 **B.Sc. in Physics**
Saurashtra University, India

Research and Academic Positions

Visiting Researcher, Hadron Physics Theory Lab., SPU Oct 2023 – Jun 2024

– Study on neutron stars, strong-field gravity and compact objects.
– Developed relativistic stellar models and investigated spacetime curvature of compact stars.

Project Student, International Centre for Cosmology, CHARUSAT May 2019 – Jul 2019
– Summer research project on General Relativity and its applications to astrophysics.
– Mentors: Dr. Dipanjan Dey and Prof. Pankaj S. Joshi.

Visiting Lecturer, B. G. Patel College of Physiotherapy, SPU, Anand Aug 2017 – Present
– Teaching Biomedical Physics, Modern Physics and Physics Laboratory.

Visiting Lecturer, Bhaikaka University, Anand Nov 2024 – Present
– Teaching Biomedical Physics, Modern Physics and Physics Laboratory.

Visiting Lecturer, B. N. Patel College of Physiotherapy, Anand Mar 2023 – Oct 2025
– Teaching Biomedical Physics, Modern Physics and Physics Laboratory.

Lecturer, Adarsh Science College, SPU, Anand May 2018 – Jul 2019
– Courses taught: Electricity and Magnetism, Heat and Thermodynamics, Optics, Classical Mechanics, Modern Physics and Physics Laboratory.

Student Supervision

- **Bhanu Kiran** (Indian Academy of Sciences Summer Research Fellow, ICTS, India)–2018
  Project: *Relativistic Compact Stellar Bodies: A Geometrical Study*
  Co-supervisor with Prof. P. C. Vinodkumar.
- **Harsh Solanki** and **Harsh Chatwani** (M.Sc. Dissertation, Sardar Patel University)–2023
  Dissertation: *The Influence of Black Holes on Light Ray and Particle Trajectories*
- **Bhavik Girnara** (M.Sc. Dissertation, Sardar Patel University)–2023
  Dissertation: *Moment of Inertia of Rapidly Rotating Neutron Stars*

Publications

Peer-reviewed journal articles: 5 **First-author papers:** 3

[1] **A. C. Khunt**, K. Yavuz Eksi, and P. C. Vinodkumar, *Impact of Anisotropy on Neutron Star Structure and Curvature*, JCAP **05**, 009 (2026).

[2] Sagar V. Soni, **A. C. Khunt**, Farook Rahaman, and A. H. Hasmani, *Energy Conditions and Stability of Charged Wormholes in $f(R, L)$ Gravity: A Comparative Analysis with Compact Objects*, Int. J. Mod. Phy. A **41**, 14 (2026).

[3] S. V. Soni, **A. C. Khunt**, and A. H. Hasmani, *A Study of Morris–Thorne Wormhole in Einstein–Cartan Theory*, Int. J. Geom. Methods Mod. Phys. **21**, 2450115 (2024).

[4] **A. C. Khunt**, V. O. Thomas, and P. C. Vinodkumar, *Relativistic Stellar Modeling with Perfect Fluid Core and Anisotropic Envelope Fluid*, Indian J. Phys. **97**, 3379–3393 (2023).

[5] **A. C. Khunt**, V. O. Thomas, and P. C. Vinodkumar, *Distinct Classes of Compact Stars Based on Geometrically Deduced Equations of State*, Int. J. Mod. Phys. D. **30**, 2150029 (2021).

Conference Proceedings

[1] **A. C. Khunt**, V. O. Thomas, and P. C. Vinodkumar, *The Properties of Massive Compact Stars Using a Geometrically Deduced Equation of State*, Proceedings of the DAE-BRNS Symposium on Nuclear Physics, Vol. 64 (2019).

[2] Abhirami R. Sarma, **A. C. Khunt**, and Vikash Kumar Ojha, *Study of Neutron Star Using NL3 Model and Dark Matter Extensions*, Proceedings of the 69th DAE Symposium on Nuclear Physics, Vol. 69, pp. 821–822 (2026).

Conferences, Workshops and Schools

GeomGravX, University of Tartu, Estonia Jun–Jul 2026
Conference Website

Foundations of GWs: Theory, Sources and Detection, IUCAA, Pune May–Jun 2026
Advanced school on gravitational-wave theory and detection. Link

Academic Visit, IUCAA, Pune Nov 2025
Research visit.

Academic Visit, IUCAA, Pune Aug 2025
Research visit.

Testing Aspects of General Relativity-IV, IIT Gandhinagar Jul 2025
Workshop Website

3rd IAGRG School on Gravitation and Cosmology, ICTS Bengaluru Oct 2024
School Website

Prof. P. C. Vaidya National Conference on Mathematical Sciences, SPU Mar 2022
Contributed Talk.

IAGRG Online Conference Dec 2020
IIT Gandhinagar, India. Conference Wensite

IWARA 2020-Video Conference Sep 2020
Mexico City (Online) Conference website.

Exposure Series in Astrophysics and Cosmology Jul 2020
Stellar Universe and SciRox Science Club, GNDU.

64th **DAE-BRNS Symposium on Nuclear Physics, Lucknow** Dec 2019
Presented a Poster.

International Workshop on Astrophysics and Cosmology, ICC-CHARUSAT Dec 2019
Participated.

Summer School, International Centre for Cosmology (ICC), CHARUSAT May–Jul 2019
Research training in General Relativity and Astrophysics.

XXXI Gujarat Science Congress (GERMI-PDPU) Feb 2017
Presented a Poster.

RAFTCT National Symposium, Nirma University Sep 2016
Presented a Poster.

Languages

Gujarati	Native
English	Professional proficiency
Hindi	Professional proficiency

Computational & Numerical Skills

Programming Languages	`Python`, `Fortran`
Mathematical Software	`Mathematica`, `Maple`, `SageMath`
Scientific Packages/codes	`Bilby` (Bayesian gravitational-wave parameter estimation), `xAct` (tensor algebra and differential geometry), `GRTensor II/III` (General Relativity), `RNS` (Rapidly rotating neutron star)
Scientific Libraries	`NumPy`, `SciPy`, `Matplotlib`, `SymPy`
Data Analysis & Visualization	`OriginLab` (scientific plotting and data analysis)
Typesetting	LaTeX, BibTeX

Commission of Trust

- Peer Reviewer, Canadian Journal of Physics (**Can. J. Phys.**)

Professional Memberships

- Lifetime Member, Indian Association for General Relativity and Gravitation (IAGRG), India.

References

Prof. P. C. Vinodkumar
Ph.D. Supervisor
Professor
Charotar University of Science and Technology, India
Email: pcvinodkumar.as@charusat.ac.in

Prof. K. Yavuz Ekşi
Research Collaborator
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Istanbul Technical University, Turkey
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Sardar Patel University, India
Email: jnpandya-phy@spuvvn.edu

Prof. A. H. Hasmani
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Sardar Patel University, India
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The Maharaja Sayajirao University of Baroda, India
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Academic Profiles

INSPIRE-HEP	INSPIRE-HEP Profile
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